

MOST HAPPENING, FACT-FILLED CURRENT AFFAIRS FROM AROUND THE WORLD

3D printers for astronauts

Astronauts need 3D printers, despite their not-so-luxurious lifestyles. You must be wondering about the need of a 3D printer in an astronaut's life. Well, there is a relevant answer too. Stuff breaks in space and replacement of the same is not an easy task while someone is in the space. So, to help those folks, Made in Space has designed a 3D printer that sidesteps Earth's gravity, when it is used in the orbit. The 3D printer, known as the Zero-G printer, is not made of molten filament but its surface tension holds a widget. There are plans to make a gizmo that would allow astronauts to melt tools. Made in Space has focused on cost-effective measures.

The team is also planning 3D-printing robots, which will be sent to Mars or the Moon. Several tests have been performed on parabolic plane flights by Made in Space that says 15 to 20 minutes are required for complete parts to be printed. The printer has been designed to be operated from the ground, most of the time.

Underwater robot to ensure ships' security

Sampriti Bhattacharyya, Massachusetts Institute of Technology graduate student in mechanical engineering, along with Professor Harry Asada, has developed an underwater robot, which looks like a football. It is aimed to inspect ships' bottom for illegal goods through an ultrasound scan.



Underwater robot



Astronauts at Made in Space

The robot was initially targeted to inspect cracks in nuclear reactors and water tanks and now it has been modified to inspect ships for smuggling activities. The working prototype of the robot was presented at the International Conference on Intelligent Robots and Systems and a group of such robots is able to stay hidden in water to conduct the inspection of ships, which arrive at ports.

The robot is oval-shaped and a little smaller than a football with a flattened panel on one side. Because of the robot's unusual design, the team used 3D printers for the structural components of the robot. Its flattened panel is waterproof so that the electrical components are protected. The other half is a spongy one, which has the propulsion system of the robot. It comprises six pumps that expel water through rubber tubes.

In the prototype, rechargeable lithium batteries have been used which last about 40 minutes. The next prototype, according to Bhattacharyya, will feature wireless rechargeable batteries and some modifications will take place in the propulsion system too.

The research was funded by National Science Foundation.

Google working on a shape-shifting TV

Google has been reportedly working on a new modular concept of a TV screen that will allow the user to rearrange the screen into different shapes and sizes, as per *The Wall Street Journal*. Now our TV screen options would not remain confined to the flat or the curved screens, we now might have something more refined to opt for.

The report has further elaborated that Google is working to develop a display comprising smaller screens, which will thereafter merge together like Legos to develop a single screen. The small pieces would facilitate the user to change the size and shape of the screen as per the requirement. The display sizes are purportedly considered as 'giant' enough to cover the wall of a living room.

The project is right now only at the planning stage, wherein Google is still trying to figure out how it would allow the picture to appear flawless after the modular parts will come together. They are trying hard to remove the borders amidst each module to have